Shiv Nadar University, Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 20/07/2026 (Week 2) **Continuous Lab Evaluation-2** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Topics: 8051 Tool Chain setup – Development Board – AT89S52

Activity:

Compile a minimal bare-metal C program, program the device, and control an LED by directly accessing the microcontroller's memory-mapped GPIO registers

Strictly No to using **Arduino APIs, HAL libraries, or RTOS services!**

Supporting Snippets:

1. Board Schematics 2. Pin diagram of 8051

Perform the following:

1. **Hardware Board Bring-Up**
   - Identify the microcontroller, debugger/programmer, power source, user LED, and relevant GPIO pin.
   - Verify the board's power rails using a digital multimeter.
   - Identify the LED GPIO port and pin from the datasheet/schematic.
   - Establish communication between the host computer and the target MCU.

2. **Bare-Metal LED Blink**
   - Configure the LED GPIO as an output using direct register access.
   - Write a delay mechanism using a simple software loop.
   - Blink the LED at approximately 1 Hz.

   Recommended Deliverables

   Students should submit:

   Hardware Bring-Up Sheet

Parameter	Student Entry
MCU	
CPU Core	
Operating Voltage	
Debug Interface	
LED GPIO Port	
LED GPIO Pin	
LED Active State	
LED Frequency (Hz)	

3. Develop a **customed library function for delay functionality**.

Deliverable form: Git repo :